

Week 1 — Prompt Engineering & ReAct

Zynxis Agentic AI Internship | Muhammad Faozan Mujtaba | Chain-of-Thought & ReAct Pattern

Chain-of-Thought (CoT)

"Let's think step by step"

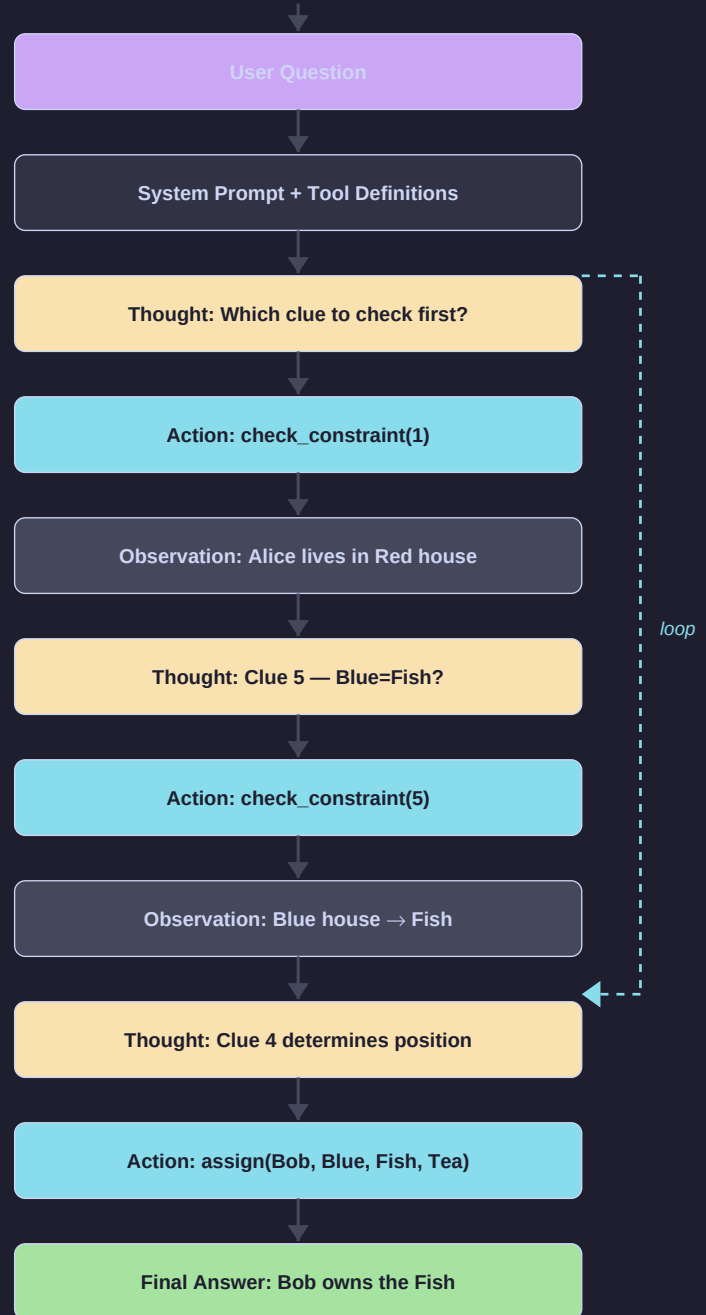


◆ *Linear reasoning chain* ◆

No external tools · One-shot output

ReAct (Reason + Act)

Thought → Action → Observation loop



◆ *Iterative reasoning with tool use* ◆

Multi-step · External tools · Observable

CoT vs ReAct — Comparison

Zynxis Agentic AI Internship | Muhammad Faozan Mujtaba | Week 1 Summary

Property	Chain-of-Thought (CoT)	ReAct (Reason + Act)
Reasoning Style	Linear, single-pass	Iterative, multi-step loop
Tool Use	None — LLM only	Yes — APIs, DBs, functions
Output	Generated entirely by LLM	Interleaved with observations
Traceability	Visible in reasoning text	Explicit per-step logs
Best For	Self-contained logic puzzles	Tasks needing external data
Failure Mode	Hallucination mid-chain	Tool call errors / bad parsing
Week 1 Use	Solve Zebra puzzle directly	Verify each clue via tools

ReAct Step Colour Legend

Thought	Internal reasoning step — LLM decides next action
Action	Tool call dispatched to the environment
Observation	Result returned from the tool
Final Answer	Agent's concluded response
System Prompt	Initial instruction context

Key Insight

ReAct = CoT + Tool Use. The agent still reasons step-by-step (like CoT), but can now interact with the world between each reasoning step. This makes it suitable for tasks that require real-time data, verification, or multi-system coordination — the foundation of every agent system built in Weeks 2 through 8 of this internship.